Triple integrals

Let V be a volume bounded by a surface S. The triple integral of $f(x,y,z)$ over the region V is defined as

$$\iiint_V f(x,y,z)\,dV = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i,y_i,z_i)\,\Delta V_i$$

By drawing lines parallel to co-ordinate axes $\Delta V_i = \Delta x_i \cdot \Delta y_i \cdot \Delta z_i$ and $dV = dx\,dy\,dz$

$$\therefore \iiint_V f(x,y,z)\,dV = \iiint_V f(x,y,z)\,dz\,dy\,dx$$

Suppose V is bounded by surfaces $z=f_1(x,y)$ and $z=f_2(x,y)$ where $f_1(x,y)$ and $f_2(x,y)$ are defined in a region R of xy plane, then

$$\iiint_V f(x,y,z)\,dx\,dy\,dz = \iint_R \left[\int_{z=f_1(x,y)}^{z=f_2(x,y)} f(x,y,z)\right] dy\,dx$$

Note volume of a closed and bounded region in space is given by $\iiint_V dV$

Problems

(1) $\int_0^1\int_0^1\int_0^1 (x^2+y^2+z^2)\,dz\,dy\,dx$

$$= \int_0^1\int_0^1 \left[\int_0^1 (x^2+y^2+z^2)\,dz\right] dy\,dx$$

$$= \int_0^1\int_0^1 \left(x^2z+y^2z+\frac{z^3}{3}\right)_0^1 dy\,dx$$

$$= \int_0^1\int_0^1 \left(x^2+y^2+\frac{1}{3}\right) dy\,dx$$

$$= \int_0^1 \left(x^2 y + \frac{y^3}{3} + \frac{y}{3}\right)_0^1 dx$$

$$= \int_0^1 \left(x^2 + \frac{1}{3} + \frac{1}{3}\right) dx$$

$$= \int_0^1 \left(x^2 + \frac{2}{3}\right) dx = \left(\frac{x^3}{3} + \frac{2}{3}x\right)_0^1 = \frac{1}{3} + \frac{2}{3} = \underline{\underline{1}}$$

(2) Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx$

~~evaluate~~ $I = \int_0^1 \int_0^{1-x} \left[\int_0^{1-x-y} z \, dz \right] dy \, dx$

$$= \int_0^1 \int_0^{1-x} \left(\frac{z^2}{2}\right)_0^{1-x-y} dy \, dx$$

$$= \int_0^1 \int_0^{1-x} \frac{1}{2}(1-x-y)^2 \, dy \, dx$$

$$= \int_0^1 \frac{1}{2}\left[\frac{(1-x-y)^3}{(-1)\times 3}\right]_0^{1-x} dx$$

$$= -\frac{1}{6} \int_0^1 \left[0 - (1-x)^3\right] dx$$

$$= -\frac{1}{6} \times (-1) \int_0^1 (1-x)^3 \, dx$$

$$= \frac{1}{6}\left[\frac{(1-x)^4}{(-1)\times 4}\right]_0^1 = -\frac{1}{6}\left[0 - \frac{1}{4}\right] = \underline{\underline{\frac{1}{24}}}$$

(3) Evaluate $\iiint_V \frac{dx \, dy \, dz}{\sqrt{1-x^2-y^2-z^2}}$ where V is the unit sphere with centre at origin.

Equation of the unit sphere with centre at origin is $x^2 + y^2 + z^2 = 1$

In the first octant

$0 \le z \le \sqrt{1-x^2-y^2}$, $0 \le y \le \sqrt{1-x^2}$ and $0 \le x \le 1$

$$I = 8\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}}\, dz\, dy\, dx$$

$$= 8\int_0^1 \int_0^{\sqrt{1-x^2}} \left(\int_0^{k} \frac{1}{\sqrt{k^2-z^2}}\right) dz\, dy\, dx \quad \left(\begin{array}{l} k^2 = 1-x^2-y^2 \\ k = \sqrt{1-x^2-y^2} \end{array}\right)$$

$$= 8\int_0^1 \int_0^{\sqrt{1-x^2}} \left[\sin^{-1}\left(\frac{z}{k}\right)\right]_0^k dy\, dz$$

$$= 8\int_0^1 \int_0^{\sqrt{1-x^2}} [\sin^{-1}(1) - \sin^{-1}(0)]\, dy\, dz$$

$$= 8\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{\pi}{2}\, dy\, dz = 4\pi \int_0^1 \int_0^{\sqrt{1-x^2}} dy\, dz$$

$$= 4\pi \int_0^1 (y)_0^{\sqrt{1-x^2}} dx = 4\pi \int_0^1 \sqrt{1-x^2}\, dx$$

$$= 4\pi \left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right]_0^1 = 4\pi \cdot \frac{\pi}{4} = \underline{\underline{\pi^2}}$$

(4) Find by triple integration the volume of the sphere $x^2+y^2+z^2=a^2$

Solution In the first octant $0 \le z \le \sqrt{a^2-x^2-y^2}$,

$0 \le y \le \sqrt{a^2-x^2}$, $0 \le x \le a$.

$$\text{Required volume} = 8\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dz\, dy\, dx$$

$$= 8\int_0^a \int_0^{\sqrt{a^2-x^2}} (z)_0^{\sqrt{a^2-x^2-y^2}} dy\, dx$$

$$= 8\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{a^2-x^2-y^2}\, dy\, dx$$

$$= 8\int_0^a \int_0^{K} \sqrt{K^2-y^2}\, dy\, dx$$

Put $K^2 = a^2 - x^2$

$K = \sqrt{a^2-x^2}$

$$= 8\int_0^a \left[\frac{y}{2}\sqrt{K^2-y^2} + \frac{K^2}{2}\sin^{-1}\frac{y}{K}\right]_0^K dx$$

$$= 8\int_0^a \left[0 + \frac{K^2}{2}\sin^{-1}(1)\right] dx$$

$$= 8\int_0^a \frac{a^2-x^2}{2}\cdot\frac{\pi}{2}\, dx$$

$$= 2\pi\int_0^a (a^2-x^2)\, dx$$

$$= 2\pi\left(a^2x - \frac{x^3}{3}\right)_0^a = 2\pi\left(a^3 - \frac{a^3}{3}\right) = \frac{4}{3}\pi a^3$$

Evaluate $\int_0^1\int_0^{1-x}\int_0^{1-x-y} (x+y+z)^2\, dx\, dy\, dz$

$$I = \int_0^1\int_0^{1-x}\int_0^{1-x-y} (x+y+z)^2\, dz\, dy\, dx$$

$$= \int_0^1\int_0^{1-x}\left[\frac{(x+y+z)^3}{3}\right]_0^{1-x-y} dy\, dx$$

$$= \int_0^1\int_0^{1-x}\left[\left(\frac{1}{3}\right)^3 - \frac{(x+y)^3}{3}\right] dy\, dx$$

$$= \int_0^1\left[\frac{y}{3} - \frac{(x+y)^4}{12}\right]_0^{1-x} dx$$

$$= \int_0^1\left[\left(\frac{1-x}{3} - \frac{1}{12}\right) + \left(0 + \frac{x^4}{12}\right)\right] dx$$

$$= \left[\frac{(1-x)^2}{6} - \frac{x}{12} + \frac{x^5}{60} \right]_0^1$$

$$= 0 - \frac{1}{12} + \frac{1}{60} + \frac{1}{6} = \frac{-5+1+10}{60} = \frac{1}{10}$$

• Evaluate $\iiint\limits_V \frac{dx\,dy\,dz}{(x+y+z+1)^3}$ where V is the volume bounded by the planes $x=0$, $y=0$, $z=0$ and $x+y+z=1$

$$I = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (x+y+z+1)^{-3}\, dz\, dy\, dx$$

$$= \int_0^1 \int_0^{1-x} \left[\frac{(x+y+z+1)^{-2}}{-2} \right]_0^{1-x-y} dy\, dx$$

$$= \int_0^1 \int_0^{1-x} \left[\frac{2^{-2}}{-2} - \frac{(x+y+1)^{-2}}{-2} \right] dy\, dx$$

$$= \int_0^1 \int_0^{1-x} \left[\frac{(x+y+1)^{-2}}{2} - \frac{1}{8} \right] dy\, dx$$

$$= \int_0^1 \int_0^{1-x} \left[\frac{(x+y+1)^{-2}}{2} - \frac{1}{8} \right] dy\, dx$$

$$= \int_0^1 \left[\frac{(x+y+1)^{-1}}{-2} - \frac{y}{8} \right]_0^{1-x} dx$$

$$= \int_0^1 \left[\frac{2^{-1}}{-2} - \frac{(1-x)}{8} - \left(\frac{(x+1)^{-1}}{-2} - 0 \right) \right] dx$$

$$= \int_0^1 \left[-\frac{1}{4} - \frac{(1-x)}{8} + \frac{(x+1)^{-1}}{2} \right] dx$$

$$= \left[-\frac{x}{4} - \frac{x}{8} + \frac{x^2}{16} + \frac{1}{2} \log(x+1) \right]_0^1 = -\frac{3}{8} + \frac{1}{2} \log 2 + \frac{1}{16}$$

$$= \frac{1}{2} \log 2 - \frac{5}{16}$$